V. Quantum Error Correction

1. Introduction

a) Setting & Problem

- Coupling to environment induces errors (i.e., uncontrolled scattering).
- Classical computers: information stored in "macroscopic" properties → errors unlikely.
- Quantum computers:
  - need qubits = "simple" quantum systems, and must store general superposition, not just $|0\rangle$ and $|1\rangle$
    → fragile!
  - should be well isolated to protect qubits, but also need coupling to "environment" (experimental apparatus) to control the computation (gates, measurements).

Q: Can we protect quantum information from noise?

Classical error correction:

Copy information, e.g. encode 1 bit in 3 bits:

$$0 \longmapsto \hat{0} := 000$$

$$1 \longmapsto \hat{1} := 111$$

"encoding"

Error model: Bit flip w/ some (small) probability p (independently on all bits):

$\Rightarrow$ typically 0 or 1 bits flipped.

Error correction ("decoding") by majority vote:

$$000,\ 001,\ 010,\ 100 \longmapsto 000$$

$$111,\ 110,\ 101,\ 011 \longmapsto 111$$

Probability for a "logical error" (i.e. on encoded bit):

$$p_{\text{error}} = \text{prob}(\geq 2 \text{ flips}) = p^3 + 3p^2(1-p)$$

$$= 3p^2 - 2p^3 < p \quad \text{for } p < 1/2.$$

↑ error quadratically suppressed!

$\Rightarrow$ effective error probability decreased.

Can be improved by:

- using more bits: $0 \mapsto 00\ldots0$, $1 \mapsto 11\ldots1$
- using ("concatenating") codes
- using smarter codes (i.e. encode several bits at once)

Quantum Error Correction:

Several potential problems when trying to generalize classical error correction codes:

- cannot copy qubits
- even if we could: what would be the "majority vote"?
- different types of errors exist,

  e.g. X (bit flip)

  or Z ("phase flip")

- errors can be continuous: there is an infinity of errors!
- measuring qubits destroys quantum information!

6) The 3-qubit bit flip code

Copy qubits in computational basis:

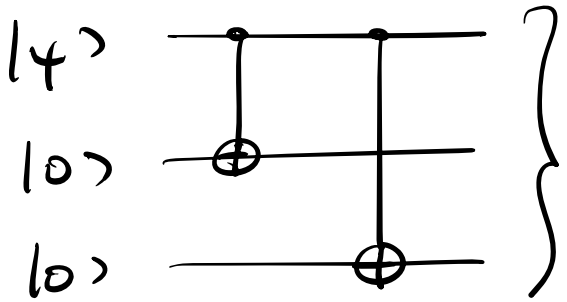
$$|0\rangle \longmapsto |\hat{0}\rangle = |000\rangle$$

$$|1\rangle \longmapsto |\hat{1}\rangle = |111\rangle$$

i.e., the encoding is a linear map

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \xmapsto{\text{encoding}} \alpha|000\rangle + \beta|111\rangle$$

Possible encoding circuit:



$$|\hat{\psi}\rangle = \alpha|000\rangle + \beta|111\rangle$$

Now consider bit flip error on qubit i:

$$|\hat{\psi}\rangle \xmapsto{\text{error}} X_i|\hat{\psi}\rangle$$

Can we correct for one bit flip error on an unknown qubit i?

Problem: Measuring the qubits in comp. basis reveals it, but also destroys superposition!

⇒ Need a measurement which only returns information about position i of error – indep. of encoded state $|\psi\rangle$!

Define "syndrome measurement" with outcomes 0, 1, 2, 3, and projectors:

0 = "no flip": $P_0 = |000\rangle\langle 000| + |111\rangle\langle 111|$

1 = "1st qubit flipped": $P_1 = |100\rangle\langle 100| + |011\rangle\langle 011|$

2 = "2nd qubit flipped": $P_2 = |010\rangle\langle 010| + |101\rangle\langle 101|$

3 = "3rd qubit flipped": $P_3 = |001\rangle\langle 001| + |110\rangle\langle 110|$

(This defines a complete measurement, as $\sum P_i = I$)

The outcome is called the "error syndrome".

Measurement of $\{P_\alpha\}$ reveals only 2 bits of info.

⇒ one qubit of information untouched!

By direct inspection: The information about is the location of the bit flip, e.g.

$$\alpha|000\rangle + \beta|111\rangle \xrightarrow[\text{on qubit 2}]{\text{bit flip}} \alpha|010\rangle + \beta|101\rangle$$

$\Rightarrow$ measurement always returns P_2,

with post-measurement state

$$\alpha|010\rangle + \beta|101\rangle \xrightarrow[\text{flip qubit 2}]{\text{recovery:}} \alpha|000\rangle + \beta|111\rangle \,!$$

$\Rightarrow$ Bit flip corrected!

Works for any single bit flip in unknown location and no flip, and for all state $|\psi\rangle$

$\Rightarrow$ suppression of error $p \rightsquigarrow 3p^2 - 2p^3$, as classically.

By linearity, this also works for part of a larger entangled state:

$$\alpha|0\rangle|a\rangle + \beta|1\rangle|b\rangle \xrightarrow{\text{encode}} \alpha|000\rangle|a\rangle + \beta|111\rangle|b\rangle$$

$$\xrightarrow[X_1]{\text{error:}} \alpha|100\rangle|a\rangle + \beta|011\rangle|b\rangle \xrightarrow[\text{correct: } X_1]{\text{meas.: } P_1} \alpha|000\rangle|a\rangle + \beta|111\rangle|b\rangle$$

What about <u>continuous errors</u>, e.g.

$$|\hat{\psi}\rangle \longmapsto e^{i\vartheta X_i}|\hat{\psi}\rangle = (\cos\vartheta I + i\sin\vartheta X_i)|\hat{\psi}\rangle \;?$$

$$|\hat{\psi}\rangle = \alpha|000\rangle + \beta|111\rangle \xrightarrow[\text{e.g. } X_3]{\text{error}} \alpha(\cos\vartheta|000\rangle + i\sin\vartheta|001\rangle) + \beta(\cos\vartheta|111\rangle + i\sin\vartheta|110\rangle)$$

$$= \cos\vartheta\,(\alpha|000\rangle+\beta|111\rangle) + i\sin\vartheta\,(\alpha|001\rangle+\beta|110\rangle)$$

$\cos\vartheta\,(\alpha|000\rangle+\beta|111\rangle)$: syndrome P_0, prob.: $|\cos\vartheta|^2$

$i\sin\vartheta\,(\alpha|001\rangle+\beta|110\rangle)$: syndrome P_3, prob.: $|\sin\vartheta|^2$

<u>Syndrome measurement collapses state into:</u>

$p = \cos^2\vartheta$: result P_0,

post-meas. state $\alpha|000\rangle + \beta|111\rangle$,

$0 \equiv$ no correction : OK ✓

$p = \sin^2\vartheta$: result P_3,

post-meas. state $\alpha|001\rangle + \beta|110\rangle$,

$3 \equiv$ correction: flip bit 3:

$\Longrightarrow \alpha|000\rangle + \beta|111\rangle$: OK ✓

Measurement of error syndrome $\{P_\alpha\}$ collapses continuous error onto one of the 4 correctable discrete errors:

- measurement "digitizes" error
- sufficient to study discrete (distinguishable) errors (will be formalized later)

A different perspective on syndrome measurement & correction (the "stabilizer formalism" — more later):

$|000\rangle, |111\rangle$: +1 eigenstates of Z_1Z_2 and Z_2Z_3 ("stabilizers")

Measure Z_1Z_2 and Z_2Z_3:

Compare qubits 1&2 and 2&3

$(+1, +1)$: no error

$(-1, +1)$: qubit 1 flipped

$(+1, -1)$: qubit 3 flipped

$(-1, -1)$: qubit 2 flipped

Now formally:

encoded state $|\hat{\psi}\rangle = \alpha|000\rangle + \beta|111\rangle$:

$$\Rightarrow Z_1 Z_2 |\hat{\psi}\rangle = |\hat{\psi}\rangle \,, \; Z_2 Z_3 |\hat{\psi}\rangle = |\hat{\psi}\rangle$$

Bit flip error, e.g. X_1:

X_1 anti-commutes with $Z_1 Z_2$

$$\Rightarrow \langle\hat{\psi}| X_1 Z_1 Z_2 X_1 |\hat{\psi}\rangle = -\langle\hat{\psi}| Z_1 Z_2 |\hat{\psi}\rangle$$

$$= -1$$

Thus:

Outcome -1 for $Z_1 Z_2$ $\Leftrightarrow$ an error which anti-commutes with $Z_1 Z_2$ has occurred.

The correction operation must satisfy the same anti-commutation relations (and some further properties) $\rightarrow$ Gates!

Have focused on X errors.

But what about Z errors?

$$\alpha|000\rangle + \beta|111\rangle \xrightarrow[\text{on qubit 1}]{Z \text{ error}} \alpha|000\rangle - \beta|111\rangle$$

This is still a state in the code space

(i.e., a valid encoded state $|\hat{\psi}\rangle$)

$\Rightarrow$ error not detectable, but it has changed $|\hat{\psi}\rangle$. After decoding, the error acts as

$$\alpha|0\rangle + \beta|1\rangle \longmapsto \alpha|0\rangle - \beta|1\rangle,$$

i.e. as a logical Z operator.

"logical operator" = operator on encoded qubit.

$\Rightarrow$ 3-qubit bit flip code cannot protect against single "phase flip error" Z.

Stabilizer picture:

Error Z_i commutes with stabilizers Z_1Z_2 & Z_2Z_3

$\Rightarrow$ it cannot be detected.

But: Z_i cannot be expressed as a product of the stabilizers Z_1Z_2 & Z_2Z_3 $\Rightarrow$ logical error!

c) The 3-qubit phase-flip code

Can we correct against Z errors?

$$Z|+\rangle = |-\rangle, \quad Z|-\rangle = |+\rangle$$

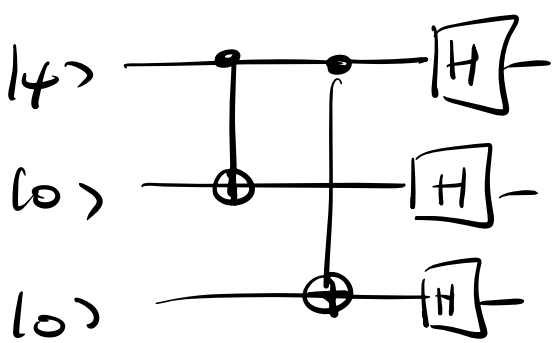
$\Rightarrow$ Z error $\hat{=}$ bit flip error in $|\pm\rangle$-basis.

Use encoding $\alpha|0\rangle + \beta|1\rangle \longmapsto \alpha|\hat{0}\rangle + \beta|\hat{1}\rangle$,

with $|\hat{0}\rangle := |+++\rangle$, $|\hat{1}\rangle = |---\rangle$:

Will protect against single Z errors!

Encoding:



Syndrome measurement:

$$\tilde{P}_\alpha := H^{\otimes 3} P_\alpha H^{\otimes 3}$$

(or via stabilizers X_1X_2 & X_2X_3).

Recovery operation:

$$H X_i H = Z_i$$

(anti-com. with stabilizers).

Problem:

Now, there is no protection against bit-flip errors X_i

— and X_i acts as a logical Z operator!

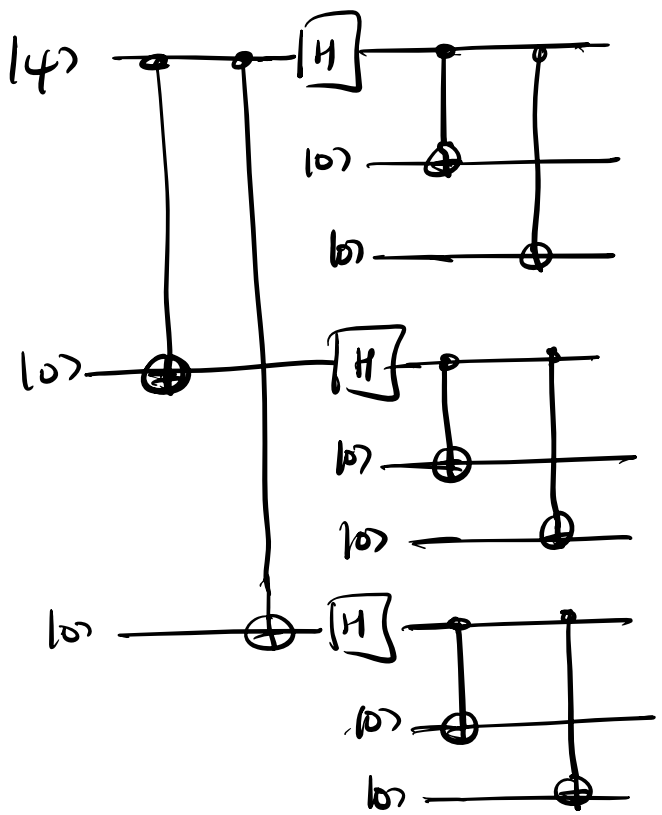
2. The 9-qubit Shor code

Solution: Concatenate (= nest) 3-qubit bit flip code and 3-qubit phase flip code:

$$|0\rangle \longmapsto |+\rangle|+\rangle|+\rangle \longmapsto \frac{(|000\rangle+|111\rangle)(|000\rangle+|111\rangle)(|000\rangle+|111\rangle)}{2\sqrt{2}}$$

$$|1\rangle \longmapsto |-\rangle|-\rangle|-\rangle \longmapsto \frac{(|000\rangle-|111\rangle)(|000\rangle-|111\rangle)(|000\rangle-|111\rangle)}{2\sqrt{2}}$$

Encoding circuit:



9-qubit Shor code

Shor code protects against arbitrary single-qubit errors!

Sufficient to focus on X_i, Z_i, and $Y \propto XZ$ errors:

Any general error $E = \alpha I + \sum \beta_i \sigma_i$ will collapse to one of those (if done right).

— More on this later! —

Intuitively:

(i) Errors X_i are corrected on "inner" layer.

(ii) Z_i error =

= logical error on qubit encoded in inner layer

= Z error on outer layer in one position

⟹ correctable!

(iii) $Y_i \propto X_i Z_i$:

Correct X_i on inner layer.

Then as in (ii): only Z error left!

More formally: Stabilizers

Code is +1 eigenstate of

$Z_1 Z_2$, $Z_2 Z_3$ ← 1st inner layer

$Z_4 Z_5$, $Z_5 Z_6$

$Z_7 Z_8$, $Z_8 Z_9$

and of ← X on intermediate qubits

$X_{(123)} X_{(456)}$ ⇓ $\alpha|000\rangle + \beta|111\rangle$

$X_{(456)} X_{(789)}$, $X_{(123)} = X_1 X_2 X_3$

⟹

$X_1 X_2 X_3 X_4 X_5 X_6$

$X_4 X_5 X_6 X_7 X_8 X_9$

These are 8 commuting operators: Measuring gives 8 bits of information ⇒ 1 qubit untouched!

Analysis of errors:

Bit flip error X_i:

e.g. X_1 anti-comm. w/ $Z_1 Z_2$

or X_2 anti-com. w/ $Z_1 Z_2$ & $Z_2 Z_3$

$\Rightarrow$ meas. of all 6 $Z_k Z_\ell$ reveals position of X_i

$\Rightarrow$ can be corrected!

Phase flip error Z_i:

E.g.: Z_1: anti-comm. w/ $X_1 X_2 X_3 X_4 X_5 X_6$

But: same holds for Z_2 or Z_3!

Yet: Z_1, Z_2 and Z_3 act identically a encoded state $|\hat{\psi}\rangle$ — can be seen by inspection, or since

$$Z_2 |\hat{\psi}\rangle = Z_1 \underbrace{(Z_2 Z_1) |\hat{\psi}\rangle}_{= |\hat{\psi}\rangle \text{ (stabilizer!)}} = Z_1 |\hat{\psi}\rangle !$$

(The 9-qubit code is a degenerate code:
different errors have the same syndrome!)

Y errors Y_i:

E.g. $Y_2 \propto Z_2 X_2$

anti-comm. w/ $Z_1 Z_2$

$Z_2 Z_3$

$X_1 X_2 X_3 X_4 X_5 X_6$

⟹ correctable e.g. via $Z_2 X_2$, or $Z_1 X_2$, ...

All single-qubit errors can be corrected!

What if errors occur on more than one qubit?

Some – but not all! – can be corrected:

e.g. $X_1 X_4$: correctable.

$Z_1 Z_2$: trivial = no error

but: $X_1 X_2$: breaks inner code

$Z_1 Z_4$: breaks outer code

3. The Quantum Error Correction Conditions

Definition: Given $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, a **Quantum Error Correction Code (QECC)** on $\mathcal{H}$ is a subspace $\mathcal{C} \subset \mathcal{H}$ (the **code space**, with $|\psi\rangle \in \mathcal{C}$ **codewords**).

We denote by $|\hat{i}\rangle$ an (arbitrary, but fixed) basis of $\mathcal{C}$.

Definition: A **noise model** on $\mathcal{H}$ is a CP map

$$\mathcal{E}(\rho) = \sum E_\alpha \rho E_\alpha^\dagger ; \quad \sum E_\alpha^\dagger E_\alpha \leq I \text{ !}$$

(i.e., error E_α occurs w/ prob. $\mathrm{tr}(E_\alpha^\dagger E_\alpha \rho)$,

e.g. $E_\alpha \propto$ tensor-product Paulis.)

Note: This is only the part of the noise which we want to correct – thus $\sum E_\alpha^\dagger E_\alpha \leq I$. The total noise is

$$\mathcal{N}(\rho) = \mathcal{E}(\rho) + \sum N_\beta \rho N_\beta^\dagger, \quad \sum E_\alpha^\dagger E_\alpha + \sum N_\beta^\dagger N_\beta = I.$$

$\leftarrow$ not correctable noise.

Definition: We say that a QECC $\mathcal{C}$ can correct for an error $\mathcal{E}$ if there exists a recovery map $\mathcal{R}$, i.e. a CP map $\mathcal{R}$ such that

$$\mathcal{R}(\mathcal{E}(\rho)) \propto \rho \qquad \forall\, \rho = |\hat{\psi}\rangle\langle\hat{\psi}|,\ |\hat{\psi}\rangle \in \mathcal{C}$$

Note: $\mathcal{R}$ must correct the error deterministically, i.e., $\mathcal{R}$ must be trace-preserving on states supported on the image of $\mathcal{C}$ under $\mathcal{E}$, i.e., on states obtained by noise from a code state.)

Theorem (Quantum Error Correction Condition):

Given $\mathcal{C}$ and $\mathcal{E}(\cdot) = \sum E_\alpha \cdot E_\alpha^\dagger$,

there exists a recovery $\mathcal{R}$ (i.e. $\mathcal{C}$ can correct for $\mathcal{E}$) if and only if

$$\boxed{\langle\hat{i}|\, E_\alpha^\dagger E_\beta\, |\hat{j}\rangle = c_{\alpha\beta}\, \delta_{ij}} \qquad (\ast)$$

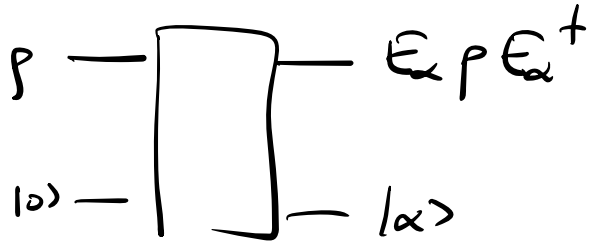
for some ONB $\{|\hat{i}\rangle\}$ $(\langle\hat{i}|\hat{j}\rangle = \delta_{ij})$ of $\mathcal{C}$.

Interpretation:

① Orthogonal states remain orthogonal (& cannot make states more orthogonal!)

② Environment learns nothing about state:

Stinespring:



$$\text{prob}(\alpha) = \left(\sum \bar{a}_i \langle \hat{i}|\right) E_\alpha^\dagger E_\alpha \left(\sum a_j |\hat{j}\rangle\right)$$

$$= \underbrace{\sum |a_i|^2}_{=1} c_{\alpha\alpha} = c_{\alpha\alpha} \text{ indep. of state.}$$

Proof:

"existence of $R \Rightarrow \otimes$":

Lemma: $\sum_\tau K_\tau |\psi\rangle\langle\psi| K_\tau^\dagger \propto |\psi\rangle\langle\psi| \quad \forall |\psi\rangle \in \mathcal{C}$

$\Rightarrow K_\tau |\psi\rangle = a_\tau |\psi\rangle$

with a_τ indep. of $|\psi\rangle$.

Proof: $\sum_\tau K_\tau |\psi\rangle\langle\psi| K_\tau^\dagger \propto |\psi\rangle\langle\psi|$

Choose any $|\chi\rangle$ s.t. $\langle\chi|\psi\rangle = 0$

$$\Rightarrow \sum_\tau \underbrace{\langle\chi|K_\tau|\psi\rangle\langle\psi|K_\tau^\dagger|\chi\rangle}_{\geq 0} \propto \langle\chi|\psi\rangle\langle\psi|\chi\rangle = 0$$

$$\Rightarrow \langle\chi|K_\tau|\psi\rangle = 0 \quad \forall \tau$$

$$\Rightarrow K_\tau|\psi\rangle = a_\tau(|\psi\rangle)\,|\psi\rangle.$$

What if $a_\tau(|\psi\rangle)$ dep. on $|\psi\rangle$? Choose $|\psi_1\rangle, |\psi_2\rangle$ s.t. $a_\tau(|\psi_1\rangle) \neq a_\tau(|\psi_2\rangle)$. Then,

$$K_\tau(|\psi_1\rangle + |\psi_2\rangle) = a_\tau(|\psi_1\rangle)|\psi_1\rangle + a_\tau(|\psi_2\rangle)|\psi_2\rangle$$

$$\neq |\psi_1\rangle + |\psi_2\rangle \quad \lightning$$

$$\Rightarrow a_\tau(|\psi\rangle) = a_\tau$$

$$\Rightarrow K_\tau|\psi\rangle = a_\tau|\psi\rangle. \qquad \blacksquare$$

Let $\mathcal{R}(\cdot) = \sum R_\gamma \cdot R_\gamma^\dagger$.

Then: $\mathcal{R}(\mathcal{E}(|\psi\rangle\langle\psi|)) \propto |\psi\rangle\langle\psi| \quad \forall |\psi\rangle \in \mathcal{C}$

Lemma

$\Longrightarrow \quad R_\gamma E_\alpha |\psi\rangle = a_{\gamma\alpha} |\psi\rangle \quad \forall |\psi\rangle \in \mathcal{C}$

ONB $|\hat{i}\rangle, |\hat{j}\rangle$:

$$\Longrightarrow \sum_\gamma \langle \hat{i} | E_\alpha^\dagger R_\gamma^\dagger R_\gamma E_\beta |\hat{j}\rangle = \sum_\gamma \overline{a_{\gamma\alpha}} a_{\gamma\beta} \langle \hat{i}|\hat{j}\rangle$$

$$=: c_{\alpha\beta}\, \delta_{ij}$$

$$\Longrightarrow \langle \hat{i} | E_\alpha^\dagger \underbrace{\Big(\sum_\gamma R_\gamma^\dagger R_\gamma \Big)}_{=\mathbb{I} \text{ on image of } \mathcal{C} \text{ under } \mathcal{E}.} E_\beta |\hat{j}\rangle = c_{\alpha\beta}\, \delta_{ij}$$

$$\Longrightarrow \quad \langle \hat{i} | E_\alpha^\dagger E_\beta |\hat{j}\rangle = c_{\alpha\beta}\, \delta_{ij} \qquad \square$$

"$\circledast \Rightarrow$ existence of R":

Construct explicit recovery channel $R(\cdot) = \sum R_\gamma \cdot R_\gamma^\dagger$.

Step 1: Use gauge degree of freedom in E_α:

$$\mathcal{E}(\rho) = \sum E_\alpha \rho E_\alpha^\dagger = \sum F_\beta \rho F_\beta^\dagger$$

if & only if $F_\beta = \sum_\alpha V_{\beta\alpha} E_\alpha$, V isometry.

Choose V unitary s.t. $\sum_{\alpha\beta} \overline{V_{\varepsilon\alpha}}\, c_{\alpha\beta} V_{\tau\beta} = \lambda_\varepsilon \delta_{\varepsilon\tau}$ diagonal

$$\langle \hat{i} | F_\varepsilon^\dagger F_\tau | \hat{j} \rangle = \sum_{\alpha,\beta} \langle \hat{i} | \overline{V_{\varepsilon\alpha}} E_\alpha^\dagger E_\beta V_{\tau\beta} | \hat{j} \rangle$$

$$= \sum_{\alpha,\beta} \overline{V_{\varepsilon\alpha}} V_{\tau\beta} \langle \hat{i} | E_\alpha^\dagger E_\beta | \hat{j} \rangle$$

$$= \sum_{\alpha\beta} \overline{V_{\varepsilon\alpha}} V_{\tau\beta} c_{\alpha\beta} \delta_{ij}$$

$$= \lambda_\varepsilon \delta_{\varepsilon\tau} \delta_{ij}$$

$\Rightarrow$ Different errors F_ε can be distinguished by a projective measurement!

Note that $\sum_\varepsilon \lambda_\varepsilon = \sum_\varepsilon \underbrace{\langle \hat{i} | F_\varepsilon^\dagger F_\varepsilon | \hat{i} \rangle}_{= \lambda_\varepsilon : \text{prob. of error } \varepsilon} \leq \langle \hat{i} | I | \hat{i} \rangle = 1.$

Step 2: Measure ε and undo error F_ε.

Want R_γ s.t. $R_\gamma F_\varepsilon | \hat{i} \rangle = \sqrt{\lambda_\varepsilon}\, \delta_{\gamma\varepsilon} | \hat{i} \rangle$! (↖ prob. of error F_ε.)

Choose $R_\gamma := \frac{1}{\sqrt{\lambda_\varepsilon}} \sum_j | \hat{j} \rangle\langle \hat{j} | F_\gamma^\dagger$.

(↖ If $\lambda_\varepsilon = 0$, then $R_\gamma = 0$ is a solution.)

$$\Rightarrow R_\gamma F_\varepsilon | \hat{i} \rangle = \frac{1}{\sqrt{\lambda_\varepsilon}} \sum_j | \hat{j} \rangle \underbrace{\langle \hat{j} | F_\gamma^\dagger F_\varepsilon | \hat{i} \rangle}_{= \lambda_\varepsilon \delta_{\gamma\varepsilon} \delta_{ij}} = \sqrt{\lambda_\varepsilon}\, \delta_{\gamma\varepsilon} | \hat{i} \rangle.$$

$$\Rightarrow R_\gamma F_\varepsilon |\hat{\psi}\rangle = \sqrt{\lambda_\varepsilon}\, \delta_{\gamma\varepsilon} |\hat{\psi}\rangle \qquad \forall |\hat{\psi}\rangle \in \mathcal{C}$$

$$\Rightarrow \mathcal{R}(\mathcal{E}(|\hat{\psi}\rangle\langle\hat{\psi}|)) = \sum_{\gamma,\varepsilon} R_\gamma F_\varepsilon |\hat{\psi}\rangle\langle\hat{\psi}| F_\varepsilon^\dagger R_\gamma^\dagger$$

$$= \sum_\varepsilon \lambda_\varepsilon |\hat{\psi}\rangle\langle\hat{\psi}| \propto |\hat{\psi}\rangle\langle\hat{\psi}| \qquad \forall |\hat{\psi}\rangle \in \mathcal{C},$$

and $\mathrm{tr}(\mathcal{R}(\mathcal{E}(|\hat{\psi}\rangle\langle\hat{\psi}|))) = \sum \lambda_\varepsilon =$

$$= \sum \langle\hat{\psi}| F_\varepsilon^\dagger F_\varepsilon |\hat{\psi}\rangle = \mathrm{tr}(\mathcal{E}(|\hat{\psi}\rangle\langle\hat{\psi}|)),$$

i.e. $\mathcal{R}$ is trace-preserving on the image of $\mathcal{C}$ under $\mathcal{E}$. ∎

Definition: *Single-qubit errors* correspond to an error model with noise operators of the form

$$E_\alpha = \sum_{k,s} \omega_{\alpha,k,s}\, \sigma_s^k$$

(σ_s^k: k'th Pauli matrix on qubit s.)

Observation: A QECC can correct for any single-qubit error if it can correct for any single-qubit Pauli error.

Proof: Code can correct for any single Pauli error $\Rightarrow$

$$\langle\hat{i}| \sigma_s^k \sigma_r^\ell |\hat{j}\rangle = c_{skr\ell}\, \delta_{ij} \Rightarrow$$

$\Rightarrow \langle\hat{i}| E_\alpha^\dagger E_\beta |\hat{j}\rangle = \tilde{c}_{\alpha\beta}\, \delta_{ij} \Rightarrow$ can correct of any single-qubit error. ∎

In particular: A QECC which can correct for single-qubit depolarizing noise

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

on any one of k qubits — i.e. a noise

$$\mathcal{E}(\rho) = (1-kp)\rho + \sum_{i=1}^{k} \frac{p}{3}(X_i\rho X_i + Y_i\rho Y_i + Z_i\rho Z_i)$$

is also robust against any single-qubit error!

Corollary: To check for robustness against arbitrary single-qubit errors, it is sufficient to check the error model with

$$\{E_a\} \propto \{I, X_1, X_2, \ldots, Y_1, Y_2, \ldots, Z_1, Z_2, \ldots\}$$

(E_a up to prefactors)

The analogous result holds for k-qubit errors vs. k-qubit Paulis.

Exercise suggestion: Check 9-qubit error correcting condition for 3-qubit & 9-qubit code!

4. Basic properties of QECCs

Focus on "binary codes":

encode k qubits in n $n > k$ qubits

Definition: The distance d of a QECC is the smallest number of Paulis $\{P_{i_k} \neq I\}_{k=1}^{d}$ s.t.

$$\langle \hat{i} | F | \hat{j} \rangle \neq \lambda \delta_{ij} \quad \text{for some } |\hat{i}\rangle, |\hat{j}\rangle \in \mathcal{C}, \; \langle \hat{i} | \hat{j} \rangle = \delta_{ij}.$$

where $F := P_{i_d} \otimes I \otimes \ldots \otimes P_{i_2} \otimes I \otimes \ldots$

(Notes: 1) For the common code classes – see later – there is also such an F which is a logical operation, i.e. $F|\hat{j}\rangle = |\hat{i}\rangle$ for $|\hat{i}\rangle \neq |\hat{j}\rangle$.

2) For classical codes, this is the smallest number of bit flips which transform one codeword to a different one.)

Notation: A binary code encoding k qubits in n qubits with distance d is denoted

$$[\![n, k, d]\!] \text{-code}$$

phys. qubits — logical qubits — distance

How many one-qubit errors can a distance-d code correct for?

Can focus on Pauli errors.

For E_α, E_β with $\leq t$ Paulis each:

$$\langle i | \underbrace{E_\alpha^\dagger E_\beta}_{\leq 2t \text{ Paulis}} | j \rangle \overset{?}{=} c_{\alpha\beta} \delta_{ij} \quad \forall E_\alpha, E_\beta$$

$$\Longleftrightarrow \quad 2t + 1 \leq d$$

Result: A distance-d code can correct t general single-qubit errors if & only if

$$\boxed{2t + 1 \leq d}$$

E.g. with a $d = 3$-code, we can correct any single-qubit error.

If the location of the error is known — that is, we additionally learn that a specific noise channel $\mathcal{E}_{\text{location}}(\cdot) = \sum \tilde{E}_\alpha \rho \tilde{E}_\alpha^\dagger$ has been applied:

$$\langle \hat{i} | \tilde{E}_\alpha^\dagger \tilde{E}_\beta | \hat{j} \rangle$$

Paulis on same location

$\Rightarrow \tilde{E}_\alpha^\dagger \tilde{E}_\beta$ has $\leq t$ Paulis

$\Rightarrow$ correctable for $\boxed{t+1 \leq d}$

Result: QECC can correct t errors on unknown locations $\Leftrightarrow$ QECC can correct $2t$ errors on known locations.

What are constraints on $[\![n,k,d]\!]$?

Definition: A code is called non-degenerate if different Pauli errors result in orthogonal

states, i.e. are distinguishable,

$$\langle \hat{j} | E_\alpha^\dagger E_\beta | \hat{i} \rangle \propto \delta_{\alpha\beta}$$

for all E_α s/ at most t ($2t+1 \leq d$) Paulis.

Theorem (Hamming bound):

For non-degenerate codes,

$$\sum_{j=0}^{t} 3^j \binom{n}{j} \leq 2^{n-k}, \quad 2t+1 = d.$$

Proof: n'a counting possibilities.

E.g.: For $k=1$, $t=1$ ($d=3$) — i.e. encodes 1 qudit, can correct for one error:

$$\underline{n \geq 5}.$$

Could there be a degenerate $[\![4,1,3]\!]$ code?

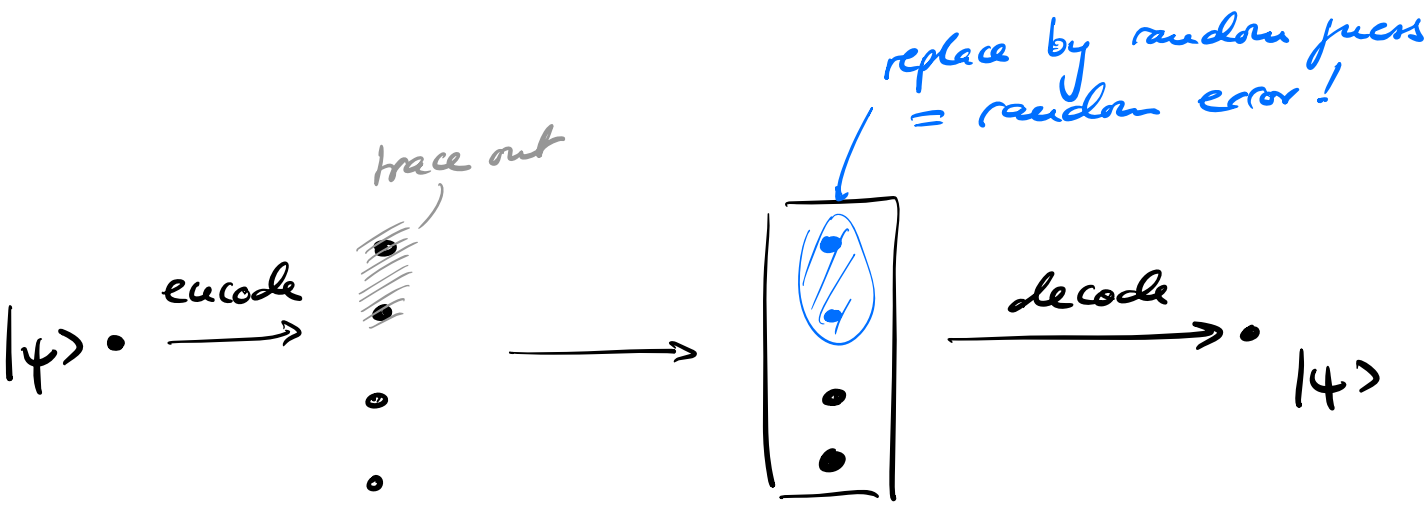
NO!

Proof:

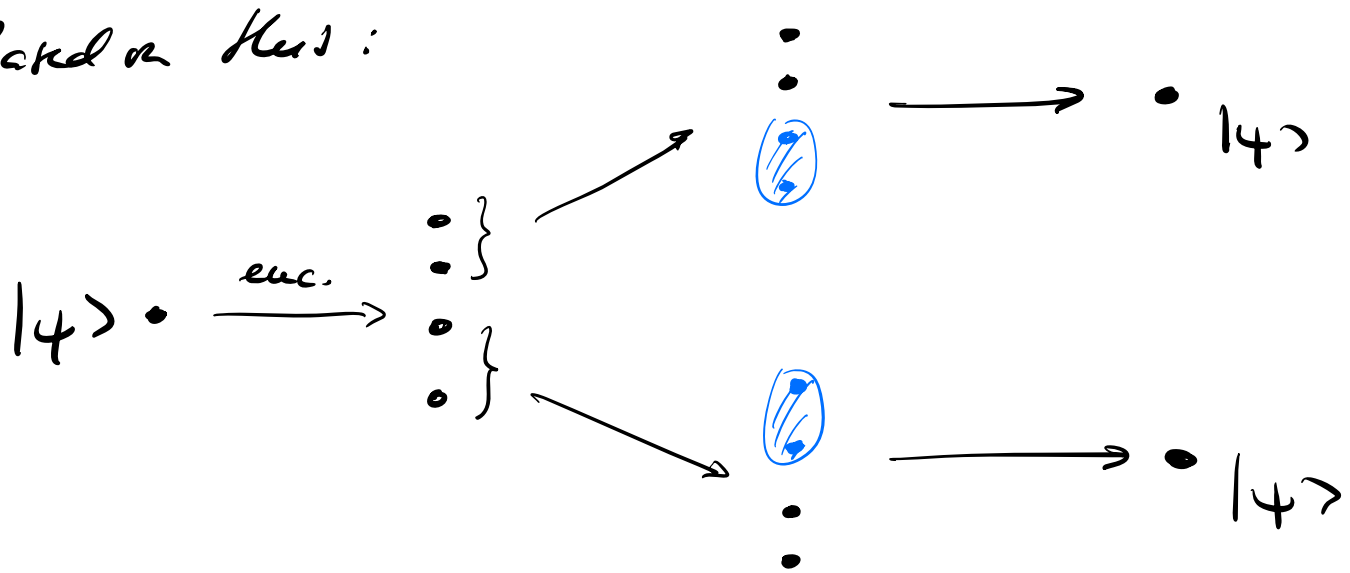
$d=3$: Can correct for maximum 1-qubit error

$\Rightarrow$ can correct for 2 erasures in known location

Can use it to recover 2 lost qubits:



Based on this:



↳ have built a quantum closet

→ No $[\![4,1,3]\!]$ code can exist,

a $[\![5,1,3]\!]$ code would be optimal!

5. Stabilizer codes

Have seen, e.g. for 5-qubit / 9-qubit code:

code space = joint +1 eigenspace of Paulis

error & correction ⟷ anti-comm. pattern

→ General framework?

a) Definition

Definition: The **Pauli group** $\mathcal{G} \equiv \mathcal{G}_n$ on n qubits is

$$\mathcal{G} := \{ i^\ell P_1 \otimes \dots \otimes P_n \mid P_i = I, X, Y, Z ; \ell = 0, \dots, 3 \}$$

Note: Any two $S_1, S_2 \in \mathcal{G}$ either **commute** or **anti-commute**.

Definition (Stabilizer group, stabilizer code)

A subgroup $\mathcal{S} \subset \mathcal{G}$ with $-I \notin \mathcal{S}$ is called a **stabilizer group $\mathcal{S}$**. Since $-I \notin \mathcal{S} \Rightarrow S_1, S_2 \in \mathcal{S}$ commute (else $S_1 S_2 S_1^{-1} S_2^{-1} = -I$); thus also implies $S = \pm \bigotimes P_i \ \forall S \in \mathcal{S}$.

The elements $S \in \mathcal{S}$ are called stabilizers.

$\mathcal{S}$ defines a subspace $\mathcal{C} \subset (\mathbb{C}^2)^{\otimes n}$,

$$\mathcal{C} := \{ |\psi\rangle \mid |\psi\rangle = S|\psi\rangle \quad \forall S \in \mathcal{S} \},$$

the code space of a stabilizer code.

$\mathcal{S}$ can also be characterized by a minimal set of generators $S_1, \dots, S_r \in \mathcal{S}$.

Lemma: $\dim \mathcal{C} = 2^{n-r}$.

Proof: (sketch!)

i) S_1 has same # of ± 1 eigenvalues (as tr $S_1 = 0$)

$\Rightarrow$ split space in half.

$\Pi_1 = \frac{1}{2}(I + S_1)$: proj. on $+1$ eigenspace of S_1.

ii) $\Pi_1 S_2 = S_2 \Pi_1$ (as $S_1 S_2 = S_2 S_1$),

and $\Pi_1 S_2 \Pi_1 = \frac{1}{2}(I + S_1) S_2$

(

+1 - eigensp. of S_2 on +1 - eigenspace of S_1

(0 on −1 - eigensp. of S_1)

$$\mathrm{tr}\left(\tfrac{1}{2}(\mathbb{I}+S_1)S_2\right) = \tfrac{1}{2}\left(\underbrace{\mathrm{tr}(S_1)}_{=0} + \underbrace{\mathrm{tr}(S_1S_2)}_{\neq 0:\ \text{such. set of gens}}\right)$$

$$= 0$$

$\Rightarrow$ S_2 has eq. # of +1/−1 eigenvals

on +1 - eigenspace of S_1

$\Rightarrow$ split again in half.

iii) continue inductively! ∎

b) Error correction conditions for stabilizer codes

What about error corr. conditions?

E_α Pauli errors.

$E_\alpha^\dagger E_\beta$ have three possibilities:

i) $E_\alpha^\dagger E_\beta$ anti-comm. with some $S \in \mathcal{S}$:

$$\langle i|E_\alpha^\dagger E_\beta|j\rangle = \langle i|E_\alpha^\dagger E_\beta \underbrace{S|j\rangle}_{S|j\rangle = |j\rangle}$$

$$= -\langle \hat{\imath} | S E_\alpha^\dagger E_\beta | \hat{\jmath} \rangle = -\langle \hat{\imath} | E_\alpha^\dagger E_\beta | \hat{\jmath} \rangle$$

$$\Rightarrow \quad \langle \hat{\imath} | E_\alpha^\dagger E_\beta | \hat{\jmath} \rangle = 0$$

$\Rightarrow$ QECC satisfied $\Rightarrow$ error correctable!

ii) $E_\alpha^\dagger E_\beta \in \mathcal{S}$:

$$\langle \hat{\imath} | \underbrace{E_\alpha^\dagger E_\beta}_{\in \mathcal{S}} | \hat{\jmath} \rangle = \langle \hat{\imath} | \hat{\jmath} \rangle = \delta_{ij}$$

$\Rightarrow$ QECC satisfied $\Rightarrow$ error correctable!

iii) $E_\alpha^\dagger E_\beta$ comm. with all $S \in \mathcal{S}$,

but $E_\alpha^\dagger E_\beta \notin \mathcal{S}$:

$\Rightarrow$ $E_\alpha^\dagger E_\beta$ acts non-trivially on code space:

it is a logical operator

In particular: $E_\alpha^\dagger E_\beta \mathcal{C} \subset \mathcal{C}$, but

$\exists |\hat{\jmath}\rangle$ s.th. $E_\alpha^\dagger E_\beta |\hat{\jmath}\rangle \neq c \cdot |\hat{\jmath}\rangle$

(else $E_\alpha^\dagger E_\beta \in \mathcal{S}$)

$\Rightarrow \langle \hat{i} | E_\alpha^\dagger E_\beta | \hat{j} \rangle \neq 0$ for $i \neq j$

$\Rightarrow$ <u>not correctable</u> ! $\lightning$ (as QEC cond. violated)

(<u>Diff. intuition</u>: Cannot tell w/ certainty if after error state is $E_\alpha |\hat{i}\rangle$ or $E_\beta |\hat{j}\rangle$, and – unlike (ii) – it does matter which of them occurred $\Rightarrow$ not correctable!)

How does the error correction work?

Correctable error model w/ errors $\{E_\alpha\}$,

E_α = product of Paulis.

Assume some error E_β (unknown!) occurred.

$$|\hat{\psi}\rangle \xmapsto{\text{error}} E_\beta |\hat{\psi}\rangle$$

Let $\sigma_i = \pm 1$ denote the commutator of S_i and E_β,

$$S_i E_\beta = E_\beta S_i \sigma_i \quad (\text{as } S_i |\hat{\psi}\rangle = |\hat{\psi}\rangle).$$

<u>Step 1</u>: Measure all S_i, $i = 1, \ldots, r$. Using the (anti) commutation, we find that the result is deterministically σ_i: $S_i E_\beta |\hat{\psi}\rangle = \sigma_i E_\beta S_i |\hat{\psi}\rangle = \sigma_i E_\beta |\hat{\psi}\rangle$

(measurement can be done using CNOTs & single-qubit rotations.)

Step 2: Pick some E_γ from $\{E_\alpha\}$ with correct commutation properties, $S_i E_\gamma = \sigma_i E_\gamma S_i$.

Step 3: Apply $E_\gamma^\dagger$ as a correction:

$$E_\beta|\hat{\psi}\rangle \xrightarrow{\text{corr.}} E_\gamma^\dagger E_\beta |\hat{\psi}\rangle.$$

Since $S_i E_\gamma^\dagger E_\beta = E_\gamma^\dagger E_\beta S_i$, we have (from (ii)) that

$E_\gamma^\dagger E_\beta |\hat{\psi}\rangle = |\hat{\psi}\rangle \Rightarrow$ error corrected.

Note: If case (iii) exists, the correction could induce a logical error!

Key question: Given a stabilizer code, what is the shortest $E_\alpha^\dagger E_\beta$ (= Pauli product) of that type (here, "short" refers to # of non-trivial Paulis) ($\to$ distance of code!)

c) Example: 3-qubit code

$$\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}$$

$$\left.\begin{aligned} S_1 &= ZZI \\ S_2 &= ZIZ \end{aligned}\right\} \Rightarrow \mathcal{S} = \{III, ZZI, ZIZ, \underbrace{IZZ}_{S_1 S_2}\}$$

$k = \underset{\|}{3} - \underset{\|}{2} = 1 \Rightarrow$ 1 encoded qubit
$\quad\;\; n \quad r$

Single-qubit X errors:

$E_\alpha = III, IIX, IXI, XII$ (up to prefactor $\sqrt{p_\alpha}$)

$E_\alpha^\dagger E_\beta = III, IIX, IXI, XII,$
XXI, XIX, IXX

$\Rightarrow$ anti-commut. w/ S_1, S_2, both $S_1 S_2$,
or an element of $\mathcal{S}$ (for III).

$\Rightarrow$ correctable!

Single-qubit Z errors:

$E_\alpha = III, IIZ, IZI, ZII$

$E_\alpha^\dagger E_\beta = ZII$ is one possibility

But: ZII commut. w/ S_1, S_2, but $\underline{ZII \notin \mathcal{S}}$!

$\Rightarrow$ Z errors not correctable!

Logical operators:

(at the same time: uncorrectable $E_a^\dagger E_b$!)

$\hat{\cdot}$ ≡ logical $\cdot$ operator

- $\hat{Z} = ZII$

  distance 1 ✓

  – or any $\hat{Z}' = \hat{Z} \cdot S$, $S \in \mathcal{S}$, e.g. IZI, ZZZ, ...

- $\hat{X} = XXX$

  – or e.g. $\hat{X}' = XXX \cdot ZZI = -YYX$, etc ...

Note: $\hat{X}\hat{Z} = -\hat{Z}\hat{X}$ — and this is all we have to require from the logical Pauli operators!

Error detection and correction:

X error E_a can be detected by anti-comm. pattern.

e.g.: • XII anti-comm. w/ ZIZ, $ZZI \in \mathcal{S}$.

• can be measured: $ZZI|\psi\rangle \stackrel{?}{=} \pm|\psi\rangle$ etc.

⇒ allows to detect error (up to a T s.t. $TS = ST \; \forall S \in \mathcal{S}$, and thus $T \in \mathcal{S}$ for

correctable errors).

d) More examples:

3-qubit phase flip code:

$$S_1 = XX\,I$$

$$S_2 = IXX$$

$$\hat{X} = XII$$

$$\hat{Z} = ZZZ$$

9-qubit Shor code:

$$S_1 = ZZI\ III\ III$$
$$S_2 = IZZ\ III\ III$$
$$S_3 = III\ ZZI\ III$$
$$S_4 = III\ IZZ\ III$$
$$S_5 = III\ III\ ZZI$$
$$S_6 = III\ III\ IZZ$$
$$S_7 = XXX\ XXX\ III$$
$$S_8 = III\ XXX\ XXX$$

8 indep. stabilizers
=
1 encoded qubit

↑ ↑ logical X of 3-qubit code!

Logical operators:

e.g.:

$\hat{Z} = ZZZ\ ZZZ\ ZZZ$

$\hat{X} = XXX\ XXX\ XXX$

— these comm. w/ S_i, as they have even # of X/Z,

but are $\notin S$, since they have odd # of X/Z.

simplest ("shortest") logical ops:

e.g. $\hat{Z} = ZII\ ZII\ ZII$

$\hat{X} = XXX\ III\ III$

($\Rightarrow$ distance 3!)

Also means that $\hat{X}$ and $\hat{Z}$ can be measured by measuring only 3 qubits!

(But: meas. a general function of $\hat{X}$ & $\hat{Z}$ requires at least 5 qubits because of no-cloning argument!)

Note: 9-qubit code is degenerate:

$E_1 = ZIIIIIIII$ and

$E_2 = IZIIIIIII$

have same syndrome, since

$E_1 E_2 = ZZIIIIIII \in S$.

e) The 5-qubit code

Consider the stabilizer code on 5 qubits w/ generators

$S_1 = XZZXI$

$S_2 = IXZZX$

$S_3 = XIXZZ$

$S_4 = ZXIXZ$

encodes $5-4 = 1$ qubit

cyclic code: $S_1, \ldots, S_5$ are cyclic permutations.

$\Rightarrow$ cyclic codewords!

$(S_5 = ZZXIX = S_1 S_2 S_3 S_4)$

Corrects any 1-qubit error:

$E_\alpha^\dagger E_\beta$ = product of ≤ 2 Paulis

$\Rightarrow$ anti-comm. w/ at least one S_i, $i = 1, \dots, 4$

(Why? Fix pos. of 1st Pauli, pick S_k which has I there. Then, 2nd Pauli must agree with that in S_k; and conversely. But: can check that those choices won't commute w/ some other S_i.)

$\Rightarrow$ correctable $\Rightarrow$ $d \geq 3$.

(And $d \leq 3$ from no-cloning: $[[5,1,3]]$-QECC!)

Error syndromes ($1 \equiv$ anti-comm. $=$ eigenval. -1)

	X error on qubit					Z error on					Y error on				
	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
S_1	0	1	1	0	0	1	0	0	1	0	1	1	1	1	0
S_2	0	0	1	1	0	0	1	0	0	1	0	1	1	1	1
S_3	0	0	0	1	1	1	0	1	0	0	1	0	1	1	1
S_4	1	0	0	0	1	0	1	0	1	0	1	1	0	1	1
S_5	1	1	0	0	0	0	0	1	0	1	1	1	1	0	1

15 errors, 15 syndromes $\Rightarrow$ non-degenerate.

All possible $2^4 - 1 = 15$ syndromes appear.

Logical operators:

$\hat{Z} = ZZZZZ$
$\hat{X} = XXXXX$

} comm. w/ all S_i (even # of X & Z in S_i), but for same reason $\notin S$!

simpler choices:

e.g. $\hat{Z}' = \hat{Z} \cdot S_3 = -YZYII$

$\hat{X}' = \hat{X} \cdot S_2 = -XIYYI$

$\Rightarrow$ distance $d = 3$

& logical info in $\hat{Z}$ or $\hat{X}$ basis can be obtained by meas. only 3 qubits!

(Note: General feature of distance-d code!)

Syndrome meas. + correction can be done using only CNOT, H, X (for corr.), and ancillas.

(Again: gen. feature of stabilizer code: need to compute parity of X & Z eigenvalues.)

§1 Epilogue: Clifford circuits & universal gate sets

Definition (Clifford group)

The Clifford group Cl_n on n qubits consists of all gates which map Paulis to Paulis:

$$Cl_n = \{ C \text{ unitary} \mid C(P_1 \otimes \dots \otimes P_n)C^\dagger = P_1' \otimes \dots \otimes P_n' \}$$

Theorem (without proof):

$$Cl_n = \{ \text{all circuits built from CNOT, } S = \begin{pmatrix} 1 & \\ & i \end{pmatrix}, H \}$$

These are also called Clifford gates and Clifford circuits.

(I.e.: Any circuit C which maps Paulis to Paulis is of this form!)

Observation: Only $T = \begin{pmatrix} 1 & \\ & e^{i\pi/4} \end{pmatrix}$ missing for a universal gate set!

Remark: Circuits consisting only of Clifford gates, w/ initial state $|0\rangle^{\otimes n}$, and Pauli measurements, can be simulated efficiently (in essence, by describing state in terms of stabilizers).

$\Rightarrow$ The "simple" gate T gives Clifford circuit universal power!

Key question for error-corrected q. computing:

How can we apply gates on encoded qubits?

Idea I: Decode $\rightarrow$ Apply $\rightarrow$ Encode:

Not good — protection lost during operation!

Idea II: Can we apply gates directly to encoded quantum information?

$\rightarrow$ Focus on **stabilizer states/codes!**

1. Clifford gates can be applied to encoded qubits:

Clifford gate $\hat{C}$ on logical qubit

$\Updownarrow$

$\hat{C}$ maps logical Paulis to logical Paulis

$\Downarrow$ logical Paulis = products of physical Paulis

$\hat{C}$ maps physical Paulis to physical Paulis

$\Downarrow$

$\hat{C}$ is a Clifford gate on physical qubits.

$\Longrightarrow$ Can implement logical $\hat{C}$ directly on physical qubits.

E.g.: $\hat{H}$ gate in 5-qubit code:

$\hat{X} = XXXXX$
$\hat{Z} = ZZZZZ$
$\hat{H}\hat{X}\hat{H} = \hat{Z}$

} need to find Clifford gate s.th. $XXXXX \leftrightarrow ZZZZZ$ & S is preserved.

2. Non-Clifford gates:

Can we also realize non-Clifford gates
— e.g. $T = \begin{pmatrix} 1 & \\ & e^{i\pi/4} \end{pmatrix}$ — in a robust way?

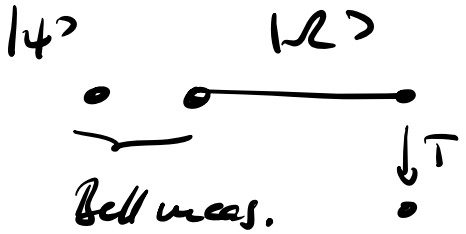
One idea: "Gate teleportation"

i) Prepare $|\chi\rangle = (I \otimes T)|\Omega\rangle$, $|\Omega\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$|\chi\rangle =$ 

"magic state"

ii) teleport state $|\psi\rangle$ of q. comp. through $|\chi\rangle$:



↳ state on the right is $T \cdot P|\psi\rangle$,
with P a Pauli matrix.

↓

Can be transformed to $T|\psi\rangle$
by Clifford gates!

- Teleportation consists only of Clifford gates and meas. in Z basis ⇒ can be done on encoded state.
- Encoded $|\chi\rangle$ can be prepared before "offline" (e.g. until success).

⇒ Can carry out universal q. computation on encoded qubits.